

Healthy Living — Vision Document

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SOEN 6481 Systems Requirements Specification — Delivery #1 — Instructor: Dr. Morales

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1. Introduction

This document sets out the vision for Healthy Living, a web-based application that helps students plan meals, keep track of their groceries, and find recipes they can actually make. It describes the problem being solved, the people the system serves, the capabilities it has to provide, and the constraints it has to work within. The focus is on what is needed and why. How the system will be built is left to the use-case model and the Software Requirements Specification.

The solution is a browser-based application with a database behind it. It keeps a record of what a student has in the pantry, lets them plan meals for a day or a week, suggests recipes they can cook with what they already own and can afford, and turns that plan into a shopping list. Around this core, the system also provides the administration, moderation, and support functions needed to run the platform safely.

1.1 References

- SOEN 6481 — Delivery #1 assignment description, Dr. Morales, Concordia University, 2026. Moodle course website.
- Vision Document template (Rational Unified Process artefact set), provided with the assignment.
- ISO/IEC/IEEE 29148:2018 — Requirements engineering. ISO/IEEE.
- Web Content Accessibility Guidelines (WCAG) 2.1, W3C Recommendation, 2018.
- Canada's Food Guide, Health Canada — nutrition reference source.
- Act respecting the protection of personal information in the private sector (Quebec Law 25). Publications Québec.

2. Positioning

2.1 Problem Statement

The problem of	unplanned eating: deciding what to eat only once you are already hungry, without knowing what is in the kitchen, what a meal will cost, or whether it is safe given your dietary restrictions
Affects	post-secondary students who prepare their own meals, and the campus health staff and families concerned with their wellbeing
The impact of which is	groceries that get bought and thrown away, money spent on expensive convenience meals, repetitive and unbalanced diets, decision fatigue, and time lost to unplanned grocery trips. Students with allergies, or with religious or ethical restrictions, also carry the extra burden of checking every ingredient by hand
A successful solution would be	a single web application that knows what the student already has, suggests recipes that fit their budget, their schedule, and their dietary restrictions, turns a weekly plan into an accurate shopping list, and warns them before

	food goes to waste, while asking for little enough effort that a busy student keeps using it
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2.2 Product Position Statement

For	post-secondary students who prepare their own meals
Who	need to eat well on a limited budget and with limited time, but who currently plan meals, track groceries, and look for recipes in three unconnected places, or do not plan at all
The Healthy Living	is a web-based meal-planning and grocery-tracking application connected to a moderated recipe database
That	links the student's pantry, meal plan, and recipe catalogue together, so that suggestions and shopping lists reflect what they actually have, what they can afford, and what they are able to eat
Unlike	recipe sites, calorie trackers, meal-kit subscriptions, and the paper lists and spreadsheets students use today, each of which only solves one part of the problem
Our product	connects inventory, plan, and purchase in one place; is free and needs no installation; and uses active moderation to keep recipes accurate and correctly labelled for allergens

3. Stakeholder Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Project Sponsor	Funds the project and owns the business case.	Approves the vision and scope; arbitrates scope, cost, and schedule; accepts each delivery.
Requirements Engineer	Elicits, analyses, and specifies the requirements.	Produces and maintains this Vision document; validates requirements with sponsor and users.
Development Team	Designs, implements, and tests the application and database.	Delivers the features of Section 5 within the constraints of Section 6; ensures the system is maintainable.
Campus Health and Food Services	Institutional partner for student nutrition programmes.	Supplies nutrition guidance; consumes anonymised usage reporting; promotes adoption.
Data Protection Officer	Guards compliance with privacy legislation.	Approves handling of personal and health-adjacent data (allergens, restrictions); sets consent and retention policy.
Recipe Contributors	Students, dietitians, and third parties who submit recipes.	Provide the content the suggestion feature depends on, with accurate ingredient and allergen information.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Student (Primary End-User)	The central persona for whom the app is built; uses the front end to manage their daily intake.	Maintains dietary profile and pantry; builds and follows meal plans; shops from the generated list; rates and reports recipes.	Self-represented
System Administrator (Admin)	Technical user responsible for backend maintenance, security, and overall health of the application.	Manages accounts and roles; monitors availability, performance, and security; performs backup and recovery.	Self-represented
Content Moderator	User responsible for overseeing the recipe database to ensure content quality and safety.	Reviews submitted recipes; corrects ingredient and allergen metadata; approves, rejects, or retires content.	Represents Recipe Contributors
Support Specialist	User who handles customer service and addresses technical or functional grievances.	Triages grievances; reproduces problems; resolves account issues; escalates confirmed defects.	Represents the Student in the defect process

3.3 User Environment

- Meal planning is a single-person task. The student plans, shops, and cooks alone, and we do not expect that to change.
- The cycle is weekly. Planning takes about five to fifteen minutes, once or twice a week, usually in the evening, and it is often interrupted. Pantry updates happen many times a week, but each one only takes a few seconds. Shopping happens once or twice a week and takes twenty to forty minutes. Planning time is always under pressure, and it is usually the first thing a student drops in a busy term.
- Environmental constraints: the shopping list gets used while standing in a grocery aisle, one-handed, on a phone, often with a weak or missing connection. Pantry updates happen in the kitchen with busy hands. Cooking is usually limited to a shared kitchen, one or two burners, and around thirty minutes on a weekday.
- Platforms in use today: modern web browsers on personal laptops, shared campus machines where students cannot install software, and phones. Native mobile apps may come later, but the browser has to remain the primary platform.
- Other applications in use: notes apps, spreadsheets, general recipe sites, and standalone calorie trackers. None of them talk to each other, so the student ends up moving information between them by hand. Healthy Living does not need to integrate with these tools, because it is meant to replace them.
- Students get no training, do not read manuals, and have no help desk to call, so the interface has to make sense on first use. The back-office users (Admin, Moderator, Support) are trained, work from desktop machines in long sessions, and can be expected to read documentation.

3.4 Key Stakeholder or User Needs

There is no existing software system; today every task below is performed by hand.

Need	Priority	Concerns	Current Solution	Proposed Solutions
Know what can be cooked now with what is on hand	High	Food that has already been bought goes to waste and the student falls back on convenience meals. There is no record anywhere of what is in the fridge.	Look in the fridge, remember, improvise; recipe sites search by name, not inventory	Pantry tracking (FE-2) feeding pantry-aware suggestion (FE-5)
Never be offered unsafe or unacceptable food	High	Allergens are a safety risk rather than a preference, and religious or ethical restrictions cannot be compromised on.	Read every ingredient list by hand, every time	Dietary profile (FE-1) enforced as a global filter on all search and suggestion
Keep food spending within budget	High	Unplanned trips and duplicate purchases quickly break a fixed budget, and the cost of a week is invisible until the student is at the checkout.	Mental arithmetic in the aisle; an abandoned spreadsheet	Cost shown per recipe and per plan (FE-3); budget stored in the profile constrains suggestion (FE-1, FE-5)
Reduce the time and effort planning takes	High	Decision fatigue. Planning gets abandoned in a busy week because it is three separate manual activities.	Paper lists, notes apps, or no plan at all	Weekly meal planner with reusable plans (FE-3)
Buy exactly what is missing	High	Forgotten items force a second trip and duplicates spoil, because the list is written from memory instead of being derived from the plan.	A handwritten or notes-app list from memory	Shopping list generated as plan minus pantry (FE-4), with checked-off items returning to the pantry
Trust that a recipe is accurate and safe	High	Wrong allergen or ingredient data can harm students, and open recipe sites are not moderated.	Cross-check recipes across several sites and hope	Moderated catalogue with a review workflow and audit trail (FE-8), fed by student reports (FE-11)
Stop groceries from spoiling unused	Medium	Money and food are wasted, since expiry is	Memory and the occasional fridge audit	Expiry tracking with use-it-first prompts (FE-6)

Need	Priority	Concerns	Current Solution	Proposed Solutions
		usually noticed only when it is too late.		
Know whether the week is balanced	Medium	Diets become repetitive, and nutrition data is entered by hand after the fact, if at all.	A calorie-tracking app, filled in manually, soon abandoned	Nutrition summaries derived automatically from the plan (FE-7)
Operate the platform securely and reliably	High (Admin)	The system holds health-related personal data, and load peaks in the evening when students plan.	None — no system exists today	Administration console with accounts, monitoring, audit logs, backup and restore (FE-9)
Resolve grievances before users abandon the product	Medium (Support)	An unresolved problem means a lost user.	None — no system exists today	In-app grievance raising with triage and escalation (FE-10)

4. Product Overview

4.1 Product Perspective

Healthy Living is a new, self-contained web application. It is not part of a larger system, and it does not replace an existing product, since these tasks are all done by hand today. It has three parts: a browser-based front end that serves all four user classes through role-specific views; a server-side back end that holds the application logic, including pantry reconciliation, plan costing, shopping-list derivation, recipe ranking, and the moderation workflow; and a relational database that stores accounts, dietary profiles, pantry inventories, meal plans, the recipe catalogue with its ingredient and allergen data, and moderation and support records.

The only external interfaces are minor ones: an email service for account verification and notifications, and a public nutrition dataset used to work out nutrition summaries. Neither of them is on the critical path. If the nutrition dataset is unavailable, planning, suggestion, and shopping-list generation all still work. The product does not connect to grocery-store, payment, learning-management, or wearable systems, and it does not need to integrate with the tools students use today, since the goal is to replace them.

4.2 Assumptions and Dependencies

Assumptions	Dependencies
Students will keep their pantry inventory up to date, as long as each update only takes a few seconds. This is the most important assumption in the document. If it turns out to be wrong, FE-4 and FE-5 lose the data they depend on and the vision would have to be revisited.	The usability of pantry entry (FE-2).
Students have regular access to a browser, and connectivity while shopping is intermittent but usable.	A mobile-usable shopping-list view that tolerates a weak connection.

Assumptions	Dependencies
There will be enough quick, low-cost, correctly labelled recipes available at launch.	Recipe contributors, and enough moderation capacity to review submissions before launch (FE-8). Suggestion quality can only be as good as the catalogue is large.
Moderation capacity keeps up with the volume of submissions. Opening the platform to community submissions without staffing the review queue would break the safety guarantee.	Content Moderator staffing; moderation tooling with filters and batch actions.
Nutrition figures taken from a public dataset are accurate enough for general guidance, and students understand that the product is informational only.	Availability and licensing of the nutrition dataset; a visible non-medical-advice disclaimer.
Personal data, including allergens and dietary restrictions, can be stored lawfully with student consent.	Compliance with Quebec Law 25 and GDPR where applicable; approval by the Data Protection Officer.
The product remains free to students and is funded as a campus service.	Hosting costs stay within budget. No advertising in the suggestion feature, because paid ranking would destroy the trust that FE-5 depends on.

5. Product Features

This section lists the high-level capabilities the system has to deliver. Each one is something a user can see, and each is written as a capability with the reason behind it rather than as a design. They are expanded further in the use-case model.

ID	Feature	Description	Priority
FE-1	Student account and dietary profile	Register, log in, and record allergens, dietary restrictions, disliked ingredients, a weekly budget, and nutrition goals. The profile works as a global filter: no recipe that violates a declared allergen or restriction is ever shown, and the student can see why a recipe was left out. This matters because allergens are a safety concern, and having to check ingredients by hand is one of the main reasons students give up on cooking.	Critical
FE-2	Grocery and pantry tracking	Record the groceries on hand (item, quantity, unit, and an optional expiry date) and add, consume, or discard them with as little effort as possible. The pantry is the system's picture of the real kitchen, and it feeds FE-4, FE-5, and FE-6. If it falls out of date, everything built on top of it becomes unreliable, so speed of entry is the main concern here.	Critical
FE-3	Meal planning	Assign recipes to meal slots across a day or a week, adjust the plan as the week changes, and reuse or	Critical

ID	Feature	Description	Priority
		duplicate an earlier plan. Total cost and nutrition are shown while the plan is being built, so the student can see the trade-offs while planning instead of at the checkout.	
FE-4	Automatic shopping-list generation	Build the list from the difference between what the plan requires and what the pantry already holds, adding up quantities across recipes and grouping items so the store can be done in a single pass. Items ticked off while shopping are added back to the pantry, so the student does not have to enter the same data twice.	Critical
FE-5	Recipe catalogue, search, and pantry-aware suggestion	Browse and search easy recipes by preparation time, cost, cuisine, and dietary compatibility. Beyond search, the system suggests recipes ranked by how many of the required ingredients the student already owns, giving extra weight to items close to expiry and staying within the profile and the budget. This is meant to answer the question students actually ask, which is what they can cook tonight rather than which recipe is the best one.	Critical
FE-6	Waste reduction and expiry awareness	Track expiry dates, highlight items that are about to spoil, suggest recipes that use them up first, and show a running total of the food and money saved.	Important
FE-7	Nutrition summary	Show nutrition information per recipe, per meal, and per day, along with how the week looks against the goals the student has set, in plain non-clinical language and with a clear statement that this is not medical advice. The point is to show an unbalanced week while there is still time to change it.	Important
FE-8	Content moderation workspace	Work through a queue of submitted and imported recipes: approve, edit, or reject them; correct ingredient, allergen, cost, and nutrition data; retire published content that turns out to be unsafe; and review an audit trail. Filters and batch actions are needed because throughput is the limiting factor for this role. No recipe reaches a student without going through this workflow, and that is what makes the guarantee in FE-1 mean anything.	Critical
FE-9	System administration and monitoring	Manage accounts and roles, monitor availability, performance, and error rates, review security and audit logs, and run backup and restore. The student never sees any of this, but it is still needed	Critical

ID	Feature	Description	Priority
		in the first release: the system holds health-related data, and a student who loses their pantry and their plans has lost the reason to use the product at all.	
FE-10	Grievance and support handling	Raise a grievance from inside the application. Support can triage, respond to, and resolve tickets using a permissioned and logged view of the affected account, escalate confirmed defects, and feed recurring issues back into the backlog.	Important
FE-11	Feedback and recipe rating	Rate a recipe, see the ratings other students have given, and report an inaccuracy such as a missing allergen or a wrong quantity. Reports go straight into the moderation queue, so students themselves help keep the content correct.	Useful

6. Other Product Requirements

Standards, platform, and environment

- Standards: WCAG 2.1 Level AA for all student-facing pages; Quebec Law 25 and GDPR where applicable (consent, minimisation, access logging, right to deletion) for personal data including allergens; OWASP Top 10 as the security baseline; national food-labelling reference quantities for nutrition figures.
- Platform: a web-based application connected to a relational database on commodity cloud infrastructure. No installed client for any user class. Clients: the two most recent major versions of mainstream desktop and mobile browsers, JavaScript enabled, no plug-ins, no specialised hardware. External interfaces: an email service and an optional public nutrition dataset.
- Environment: students use the system at home, in the library, and in grocery stores, on laptops, shared machines, and phones, with variable connectivity. The shopping list has to be readable and usable one-handed in a store aisle. Back-office users work from desktop machines on stable connections. The system has to be operable by a small team, roughly one administrator, one or two moderators, and one support specialist.

Quality ranges

Attribute	Range	Priority
Performance	Pages respond within 2 seconds, recipe suggestion within 3 seconds, and shopping-list generation within 2 seconds, at 200 concurrent sessions.	High
Scalability	At least 5,000 registered students and 200 concurrent sessions in the first release.	Medium
Availability	99% monthly. Planned maintenance outside 17:00–21:00 local time, the evening planning peak.	High
Usability	A first-time student, untrained, adds a pantry item in under 15 seconds and produces a first weekly plan in under 10 minutes.	High

Attribute	Range	Priority
Robustness	The shopping list stays usable on a degraded connection. If the nutrition dataset fails, only FE-7 is affected; planning, suggestion, and list generation carry on working.	High
Content correctness	No published recipe with incorrect allergen labelling, and every recipe passes moderation before it is published. An allergen error is a safety incident rather than an ordinary content defect.	High
Security	Credentials hashed; TLS throughout; role-based access control; every Support access to student data is logged.	High
Recoverability	Daily backups; recovery point objective 24 hours, recovery time objective 4 hours.	High
Supportability	Support tickets acknowledged within one business day; moderation queue cleared within two.	Medium

Constraints

- The product must be web-based and connected to a database; no native or installed client in this release.
- It must run on commodity cloud infrastructure within the operating budget of a free student service.
- Nutrition information must never be presented as medical or dietetic advice, and a disclaimer has to appear wherever it is shown.
- Recipe and nutrition content must be licensed to permit redistribution within the platform.
- Advertising and paid placement are excluded from recipe suggestion, because paid ranking would undermine the trust that FE-5 depends on.
- Out of scope: grocery-store ordering and payment integration, native mobile applications, wearable-device integration, and clinical dietary prescription.

Documentation requirements

- Student user guide (Medium): a short online guide covering registration, the dietary profile, pantry entry, planning, and shopping. It should stay short. If students need a long manual, the problem is with the interface and not with the documentation.
- Online help (Medium): context-sensitive inline guidance, including why a recipe was excluded by the dietary profile, and a searchable FAQ.
- Operational guides (High): deployment, monitoring, and backup procedures for the Administrator, a moderation-workflow guide for the Moderator, and a triage and escalation guide for Support. These users are trained and will read documentation.
- Labelling and packaging (Medium): consistent Healthy Living branding across all role-specific views; a visible non-medical-advice statement on every nutrition view.
- Installation (Low): no end-user installation guide is required; the product runs in a browser.

Priority of these requirements

Content correctness, security, privacy compliance, and the usability of pantry entry are all High, and none of them can be traded away in the first release. The first three decide whether the product can be operated at all, and the fourth decides whether anyone will keep using it. Performance, availability, and recoverability are also High, but they can be tuned after release. Scalability, supportability, and the documentation requirements are Medium, and they will probably change once the system is used beyond a single campus. The biggest risk of the group is content correctness, because it depends on human moderation capacity rather than on code, and a failure there is a safety problem rather than an inconvenience.

Appendix A — Effort Log (Task 0)

Date	Section / Activity	Hours	Notes
July 9, 2026	Reading the assignment; studying the RUP Vision template	0.5	Setup
July 9, 2026	Section 1 — Introduction and References	0.75	
July 9, 2026	Section 2 — Positioning	1.0	
July 10, 2026	Section 3.1–3.2 — Stakeholder and user summaries	1.0	
July 10, 2026	Section 3.3 — User environment	0.75	
July 11, 2026	Section 3.4 — Key stakeholder and user needs	1.5	Longest section
July 11, 2026	Section 4 — Product overview, assumptions and dependencies	1.0	
July 12, 2026	Section 5 — Product features	1.5	
July 12, 2026	Section 6 — Other product requirements	1.25	
July 12, 2026	Review, formatting, version control	0.75	Committed to repository
	Total	10.0	

Version control: this document is maintained in a repository so that revisions can be tracked and updated with feedback from future deliveries. Repository: <https://github.com/sachin181/soen6481-healthy-living-vision>